

Rishon Ghosh

(236) 509-8685 | Vancouver, BC | rishonghosh@icloud.com | linkedin.com/in/rishon-ghosh | github.com/rishon-g

Education

Simon Fraser University

Burnaby, BC, Canada

Bachelor of Science, Computer Science (Joint Major with Business) | **GPA: 3.89**

Expected Grad Date: Dec 2028

- **Coursework:** Algorithms & Data Structures, Introduction to Software Engineering, Business Technology Management
- **Awards:** Dean's Honour Roll, President's Honour Roll, SFU Undergraduate Open Scholarship

Experience

Software Engineer Consultant

Vancouver, BC

Self-Employed, RG Tech

Dec 2025 - Present

- Architected Python and Flask REST backends for **3 small-business clients**, turning existing sales and inventory data into reports
- Engineered SQL schemas and ETL scripts consolidating spreadsheet and POS data, saving **12 hours** of manual entry weekly
- Optimized database performance through indexing and query refactoring, reducing average API response times by **40%**
- Implemented secure authentication with password hashing and input validation to protect client and customer records
- Executed client discovery, scoping, and sprint planning to translate business needs into technical requirements delivered on schedule

Projects

TrustLine | TypeScript, React, Node.js, ElevenLabs, Gemini, Twilio, Supabase, Docker

Oct 2026

- Architected a real-time audio intelligence pipeline streaming live phone calls from Twilio Media Streams to ElevenLabs Scribe v2 Realtime over WebSockets, transcribing **8 kHz** telephony audio with no transcoding and warning callers mid-call in **5 languages**
- Engineered a dual-layer scam detection engine fusing a **~0.01 ms** rules layer across **12 scam tactics** with schema-constrained, Zod-validated LLM scoring (Google Gemini), achieving **100% precision and recall** with **0 false positives** on a 17-call labelled set at **1.1 s** median latency
- Established an evaluation and CI framework measuring precision, recall and alert latency, gating every push through GitHub Actions with **187 automated tests** across **28 suites**
- Orchestrated production deployment on **\$0** infrastructure: a Dockerized call server on Railway with health-checked, path-filtered auto-deploys, and a React/Vite frontend with serverless APIs on Vercel

FlashCart | Go, TypeScript, React, AWS Lambda, DynamoDB, EventBridge, SQS, CDK

Sep 2026

- Built a serverless flash-sale checkout in Go on AWS Lambda and API Gateway that sustained **~6,900 req/s** at **1,000** concurrent k6 users with **132 ms** average and **194 ms** p95 latency
- Prevented overselling by committing the stock decrement and order insert in one DynamoDB transaction keyed on the client's idempotency key: **0** oversold units across **145,000+** requests, backed by a **500**-goroutine race test
- Removed dual writes with a transactional outbox (DynamoDB Streams to EventBridge Pipes to SQS), an idempotent worker with compensating refunds, and a dead-letter queue with CloudWatch alarms
- Defined the stack in AWS CDK with template assertion tests; GitHub Actions runs Go tests against DynamoDB Local and deploys through keyless OIDC on every push to main

Graphite Gambit | Java, libGDX, JUnit, Mockito, Maven, JaCoCo

Jan 2026

- Led a team building a 2D top-down game in Java and libGDX, establishing weekly agile sprints and a merge-conflict protocol
- Created a map-loading system and collision engine, using mathematical proofs in testing to verify safe entity movement and bounds
- Automated a testing suite with JUnit and Mockito, achieving **71%** branch coverage across core game packages

Rubik's Cube Solver | Java

Oct 2025

- Programmed an optimal Rubik's Cube solver in Java using Iterative Deepening A* (IDA*) search, finding solutions in under **10s**
- Constructed memory-efficient pattern databases using combinatorial indexing and factorial mapping for $O(1)$ heuristic lookups
- Compressed a **43-quintillion** state space into a **~2MB** footprint, keeping heuristic data fully in memory during search

Technical Skills

Languages: Go, C++, TypeScript, Java, Python, SQL, C, JavaScript, HTML/CSS

Frameworks & Libraries: React, Node.js, Flask, gRPC, Protocol Buffers, Zod, libGDX, JUnit, Mockito

Cloud & DevOps: AWS (Lambda, API Gateway, DynamoDB, SQS, EventBridge, S3, CloudFront, CDK), Docker, GitHub Actions

Databases: DynamoDB, PostgreSQL (Supabase), SQLite

Developer Tools: Git, CMake, Maven, k6, JaCoCo, VS Code

Concepts: Distributed Systems, Concurrency, Idempotency, Event-Driven Architecture, REST APIs, CI/CD, Infrastructure as Code

Business: Financial Accounting, Excel, PowerPoint, Sprint Planning & Execution